

LUWUK BUBUNG, Indonesia

WMO: 970860

Lat: 1.0414S

Lon: 122.7711E

Elev: 7

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	74.0	74.8	62.8	85.5	78.0	64.8	91.9	78.5	18.5	79.8	16.6	80.8	5.1	280	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	90.5	79.5	89.7	79.5	89.0	79.4	81.0	88.0	80.7	87.8	80.2	87.3	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.4	86.4	78.4	147.2	86.1	78.1	145.9	85.8	44.7	88.5	44.3	88.1	43.9	87.7	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 15.9	13.7	11.9		71.8	92.6	1.6	1.4	70.6	93.6	69.7	94.4	68.8	95.2	67.7		
(5)				65.2	82.4	2.2	1.1	63.6	83.2	62.2	83.8	61.0	84.5	59.4	85.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	83.3	83.1	83.0	82.2	81.8	80.5	79.6	79.8	80.8	82.4	83.2	83.3	(6)
(7)		DBStd	2.01	1.54	1.55	1.50	1.38	1.60	1.66	1.58	1.61	1.47	1.28	1.27	1.45	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11651	1034	928	1022	967	985	914	919	923	925	1006	995	1033	(10)
(11)		CDD65	6176	569	508	557	517	520	464	454	458	475	541	545	568	(11)
(12)		CDH74	67586	6864	6000	6539	5830	5633	4482	3960	4038	4765	6162	6487	6826	(12)
(13)		CDH80	21261	2629	2204	2357	1888	1626	954	646	713	1165	2053	2408	2618	(13)
(14)	Wind	WSAvg	5.2	4.1	4.3	4.3	4.4	5.2	5.8	6.4	7.2	7.0	5.8	4.4	3.9	(14)
(15)	Precipitation	PrecAvg	57.4	3.7	4.0	5.7	6.1	5.8	8.7	7.7	4.5	2.4	1.7	2.9	4.2	(15)
(16)		PrecMax	96.0	9.1	12.2	12.1	12.8	11.3	32.3	19.4	14.7	6.5	5.0	7.6	11.2	(16)
(17)		PrecMin	28.7	0.1	0.3	1.1	2.6	2.4	3.1	3.1	0.0	0.2	0.0	0.6	1.9	(17)
(18)		PrecStd	12.5	2.1	2.4	3.0	2.2	2.1	5.7	4.1	3.4	1.9	1.7	1.6	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	91.4	90.8	89.8	89.1	87.7	86.4	86.5	87.9	89.3	90.6	91.0	(19)
(20)			MCWB	79.0	79.2	80.0	80.0	80.2	78.9	77.7	77.0	78.0	78.5	79.7	79.6	(20)
(21)		2%	DB	90.0	89.9	89.6	88.6	87.9	86.4	85.4	85.5	86.8	88.4	89.6	90.0	(21)
(22)			MCWB	79.3	79.4	79.7	79.7	79.5	78.3	77.2	76.6	77.4	78.4	79.4	79.7	(22)
(23)		5%	DB	89.1	88.9	88.7	87.6	87.0	85.6	84.4	84.6	85.9	87.7	88.8	89.1	(23)
(24)			MCWB	79.3	79.2	79.4	79.4	79.1	77.9	76.8	76.2	77.0	78.3	79.2	79.6	(24)
(25)		10%	DB	88.2	87.9	87.8	86.7	86.0	84.6	83.4	83.7	85.1	86.9	87.9	88.2	(25)
(26)			MCWB	79.2	79.0	79.2	79.1	78.7	77.5	76.3	75.7	76.6	78.0	78.9	79.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.3	81.0	81.5	81.1	80.9	80.0	79.1	78.8	79.5	80.2	81.0	81.6	(27)
(28)			MCD B	87.8	88.0	88.7	87.5	87.4	85.8	85.0	84.7	85.9	87.1	88.6	88.6	(28)
(29)		2%	WB	80.7	80.5	80.7	80.5	80.3	79.2	78.3	77.9	78.7	79.7	80.2	80.9	(29)
(30)			MCD B	87.9	88.0	87.9	87.2	86.6	85.1	84.0	84.0	85.3	86.8	87.7	88.2	(30)
(31)		5%	WB	80.1	80.0	80.0	79.9	79.7	78.6	77.5	77.1	78.0	79.1	79.8	80.3	(31)
(32)			MCD B	87.4	87.4	87.1	86.4	85.9	84.5	83.1	83.4	84.6	86.4	87.4	87.7	(32)
(33)		10%	WB	79.7	79.4	79.6	79.4	79.1	77.9	76.8	76.3	77.3	78.5	79.3	79.8	(33)
(34)			MCD B	87.0	86.8	86.6	85.7	85.1	83.6	82.3	82.5	83.8	85.9	86.8	87.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	9.6	9.5	9.1	8.4	7.6	7.3	7.6	8.5	9.4	9.8	9.9	(35)
(36)			MCDBR	10.8	10.7	10.6	10.1	9.3	8.8	8.5	8.6	9.1	9.9	10.6	11.1	(36)
(37)			MCWBR	5.9	5.9	5.6	5.5	5.6	5.5	5.5	6.0	6.6	6.7	6.1	5.7	(37)
(38)		5% WB	MCD BR	10.0	10.1	9.9	9.5	9.0	8.3	8.0	8.3	8.7	9.6	10.1	10.3	(38)
(39)			MCWBR	5.7	5.7	5.6	5.4	5.5	5.4	5.4	5.9	6.1	6.4	5.9	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.415	0.417	0.420	0.410	0.403	0.413	0.412	0.419	0.422	0.457	0.438	0.421	(40)	
(41)		taud	2.474	2.463	2.458	2.493	2.528	2.512	2.513	2.473	2.449	2.328	2.397	2.464	(41)	
(42)		Ebn at Noon	291	291	287	283	278	271	273	277	282	276	282	287	(42)	
(43)		Edh at Noon	36	37	37	35	33	32	33	35	37	42	39	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1644	1691	1697	1650	1511	1363	1384	1534	1807	1864	1753	1624	(44)	
(45)		RadStd	67	117	100	83	104	115	150	143	128	109	89	102	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page